Q1: 16 March (Shift 1) - Numerical

The resistance $\mathrm{R}=\frac{\mathrm{V}}{\mathrm{I}}$, where $\mathrm{V}=(50 \pm 2) \mathrm{V}$ and $\mathrm{I}=(20 \pm 0.2) \mathrm{A}$. The percentage error in R is 'x' \%. The value of 'x' to the nearest integer is ____

Q2: 18 March (Shift 1) - Single Correct

The time period of a simple pendulum is given by $\mathrm{T}=2 \pi \sqrt{\frac{\ell}{\mathrm{g}}}$. The measured value of the length of pendulum is 10 cm known to a 1 mm accuracy. The time for 200 oscillations of the pendulum is found to be 100 second using a clock of 1 s resolution. The percentage accuracy in the determination of 'g' using this pendulum is 'x'. The value of 'x' to the nearest integer is:-

(1) 2%

(2) 3%

(3) 5%

(4) 4%

Q3: 18 March (Shift 2) - Numerical

The radius of a sphere is measured to be $(7.50 \pm 0.85) \mathrm{cm}$. Suppose the percentage error in its volume is x. The value of x, to the nearest

x, is ___.

Answer Key

Q1 (5) **Q2** (2) **Q3** (34)

Q1 (5)

$\frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100$

$\%$ error in $R = \frac{2}{50} \times 100 + \frac{0.2}{20} \times 100$

$\%$ error in $R = 4 + 1$

$\%$ error in $R = 5\%$

Q2 (2)

$g = \frac{4\pi^2 \ell}{T^2}$

$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + 2\frac{\Delta T}{T} = \frac{0.1}{10} + 2\left(\frac{\frac{1}{200}}{0.5}\right)$

$\frac{\Delta g}{g} = \frac{1}{100} + \frac{1}{50}$

$\frac{\Delta g}{g} \times 100 = 3\%$

Q3 (34)

$\because v = \frac{4}{3}\pi r^3$

taking log & then differentiate

$\frac{dV}{V} = 3\frac{dr}{r}$

$= \frac{3 \times 0.85}{7.5} \times 100\% = 34\%$